

DTSA 5734

The Structured Query Language (SQL)

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	AI Engineer (63.5 / 100)	33.6 / 100
SPECIALIZATION Databases for Data Scientists		

34

CURRENT AVG SCORE
across 6 job roles

78

POTENTIAL SCORE
with recommended additions

+53

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

In this course students will thoroughly learn the Structured Query Language. Study includes all ANSI standard SQL commands and syntax. Lectures are supplemented with thorough hands-on lab assignments and exercises. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	63.5	99.2	+35.6	HIGH
ML Engineer	33.5	83.8	+50.2	HIGH
Machine Learning Engineer	32.1	81.0	+48.9	HIGH
Data Scientist	27.0	80.0	+53.0	HIGH
Applied Scientist	25.4	64.0	+38.6	HIGH
MLOps Engineer	19.8	60.1	+40.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Model Deployment & Serving 2 of 6 roles

production

LLMs & Generative AI 1 of 6 roles

storage

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.
- LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

63.5 → 99.2 (+35.6 pts)

data engineering models learning build form software technical

ML Engineer

33.5 → 83.8 (+50.2 pts)

data learning systems models production engineering starting build

Machine Learning Engineer

32.1 → 81.0 (+48.9 pts)

data engineering learning tools solutions business time design

Data Scientist

27.0 → 80.0 (+53.0 pts)

data product characteristics evolution simply insights different build

Applied Scientist

25.4 → 64.0 (+38.6 pts)

- storage
- problems
- models
- technical
- data
- learning
- systems
- engineering

MLOps Engineer

19.8 → 60.1 (+40.2 pts)

- data
- production
- text
- systems
- learning
- engineering
- training
- tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.